

K-FIELD

MASS-BRIDGE LOCKED PROJECTIVE RATIO SYSTEM

Cell3 Shape as a Forced Constraint

A foundational educational reference on the exact proton-electron mass ratio, inverse fine-structure ratio, Compton-classical-radius triangle, projective face-area lock, direction-preserving Cell3 correction, complete base-cell geometry, boundary planes, and 27-node grammar.

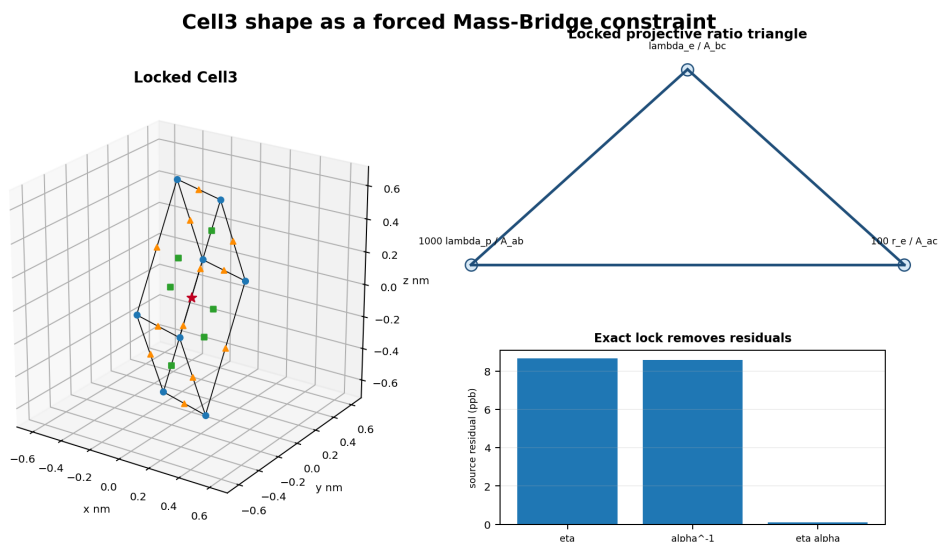


Figure 1. Title composite generated from the authoritative Cell3 basis, the attached Mass-Bridge ratios, and the exact locked constants. No illustrative or guessed geometry is used.

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Frontispiece: authoritative locked base cell, planes, and all nodes

Locked Cell3 base cell: six boundary planes and all 27 half-step nodes

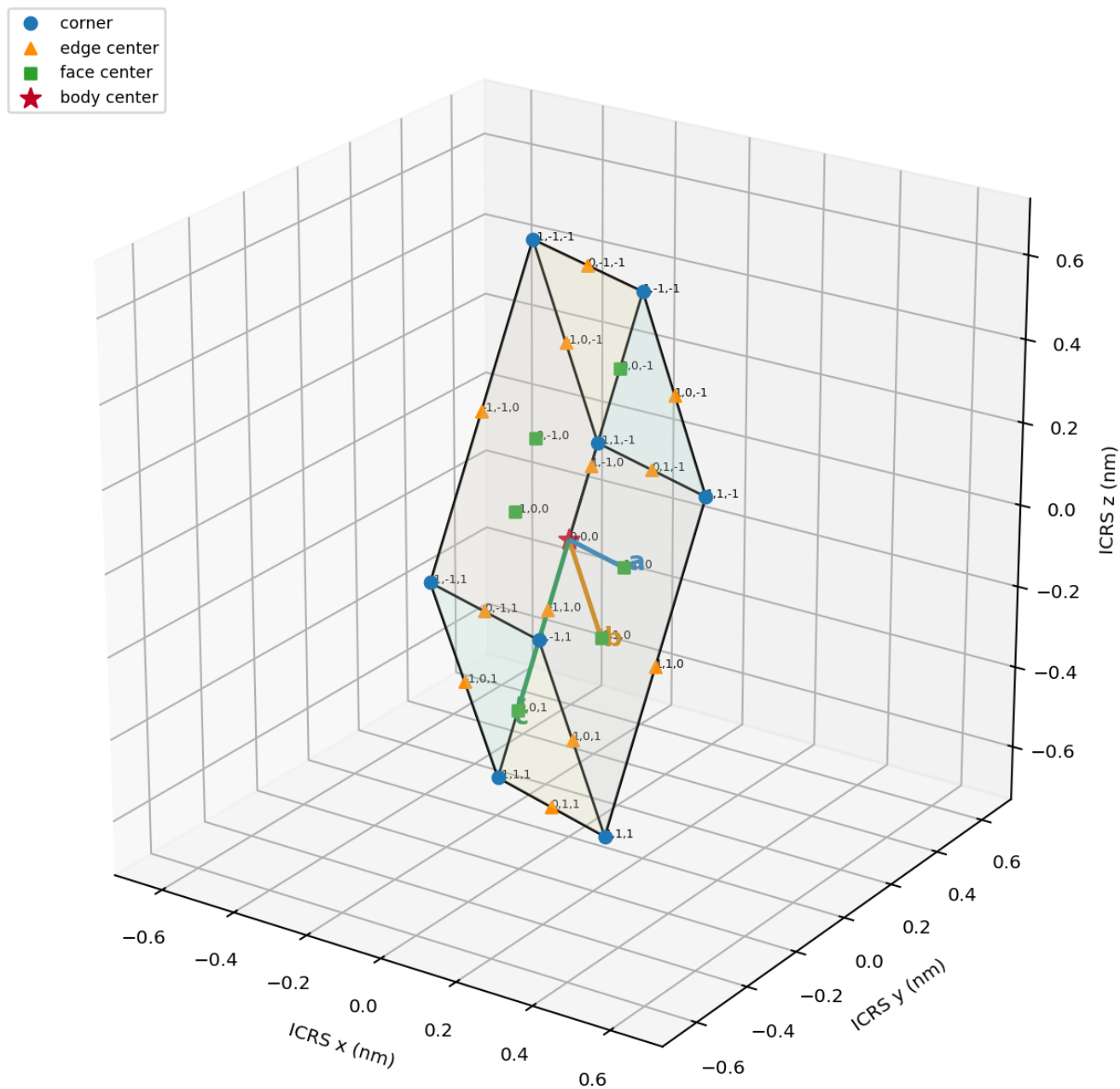


Figure 2. Dimensionally correct centered Cell3 in ICRS coordinates after the exact direction-preserving ratio lock. The figure contains all six boundary planes, twelve edges, twenty-seven half-step nodes, integer node addresses, and primitive basis directions. Axes are in nanometres and use equal three-dimensional scale.

Boundary-plane atlas

Boundary-plane atlas: complete nine-node arrays with dimensionally correct in-plane distances

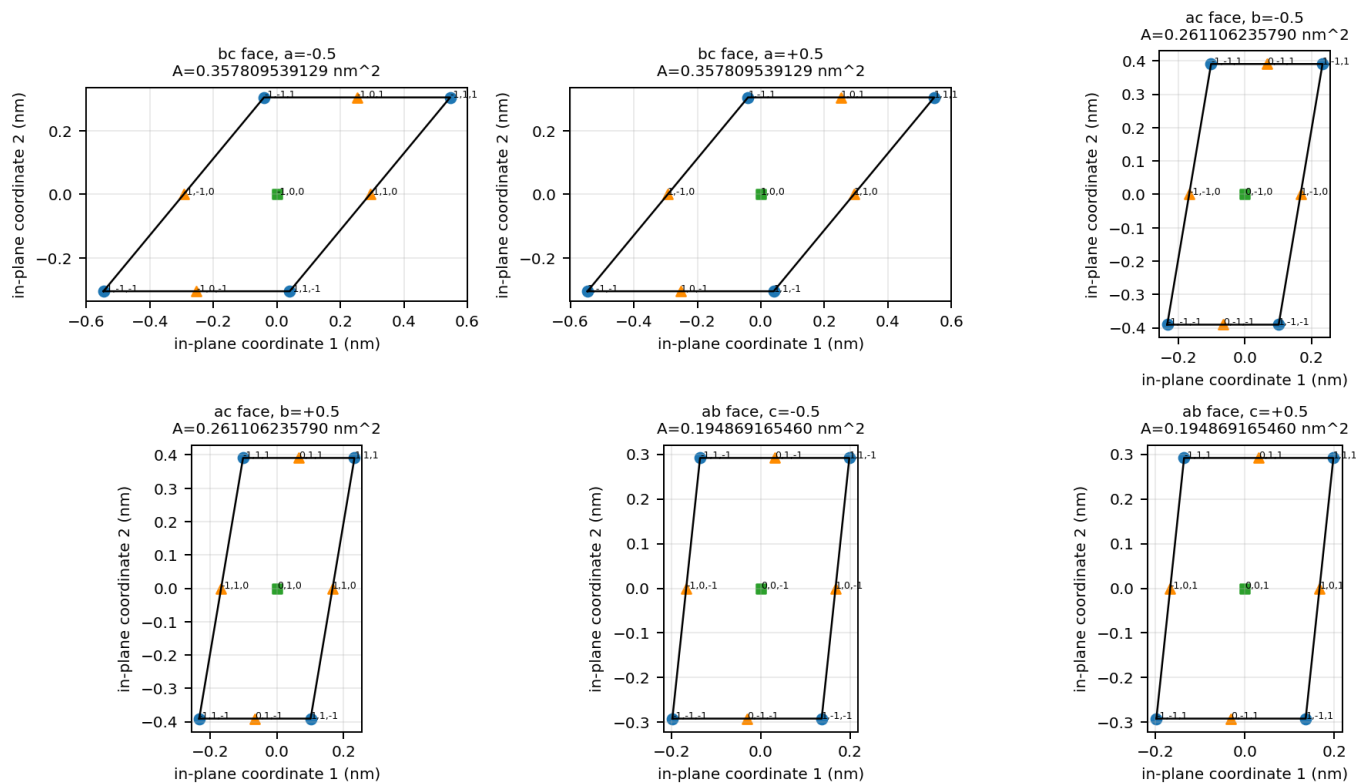


Figure 3. All six Cell3 boundary planes. Each panel preserves actual in-plane nanometre distances and shows the complete nine-node half-step array for that face, including every integer address.

Abstract

The Mass-Bridge is an exact projective ratio system connecting three Cell3 face channels to three physical length scales: the electron reduced Compton wavelength, the classical electron radius, and the proton reduced Compton wavelength. The locked identities are $1000 A_{bc}/A_{ab} = \eta$, $100 A_{bc}/A_{ac} = \alpha^{-1}$, and $10 A_{ac}/A_{ab} = \eta \alpha$. The third identity follows algebraically from the first two and eliminates the shared face A_{bc} . It is therefore an intrinsic, normalization-independent Cell3 shape invariant. The face-area triple is fixed to one point in projective area space, $[A_{bc}:A_{ac}:A_{ab}] = [1:100 \alpha:1000/\eta]$. Only a uniform geometric scale remains free. The authoritative source Cell3 already lies within about 8.6 parts per billion of the two primary ratios and within 89.1 parts per trillion of the intrinsic cross-ratio. This paper imposes the exact lock without changing the established vector directions, included angles, plane normals, node grammar, or topology. The correction is the positive diagonal map $B_{\text{locked}} = B_{\text{source}} \text{diag}(s_a, s_b, s_c)$, with edge-scale changes of only -0.042016, -0.006301, and +0.006301 parts per billion. With A_{bc} retained as the normalization anchor, the corrected basis exactly realizes the CODATA 2022 ratios. The result establishes Cell3 shape as a forced constraint within the established Cell3 construction. The combinatorial grammar fixes the 27 addresses and six face planes; the physical triangle fixes the projective face-area point; the existing orientation and angular grammar select the Euclidean realization; and the remaining degree of freedom is uniform scale. This report provides the complete derivation, numerical data, dimensionally correct figures, node and plane ledgers, reconstruction equations, and a companion machine-readable record.

Table of Contents

Frontispiece: authoritative locked base cell, planes, and all nodes	2
Boundary-plane atlas	3
Abstract	4
1. Canonical locked statement	6
2. Three physical scales form one closed triangle	6
2.1 Why the reduced Compton wavelength is the correct rotational scale	7
3. From three equations to two independent constraints	8
4. Why the Cell3 shape becomes a forced constraint	8
5. Exact direction-preserving Cell3 correction	9
5.1 Locked primitive basis	10
6. Face-area closure and numerical comparison	10
7. Base-cell planes and metric data	11
8. The 27-node grammar remains exact	11
9. Uniform scale is the only remaining geometric freedom	12
10. Consequences of the forced constraint	13

11. Reconstruction protocol	14
12. Formula ledger	14
13. Data provenance and reproducibility contract	15
Appendix A. Locked matrices	16
Locked basis B (columns are a, b, c)	16
Locked direct metric $M=B^T B$	16
Locked reciprocal metric $R=M^{-1}$	16
Appendix B. Complete boundary-plane ledger	16
Appendix C. Complete 27-node locked Cell3 ledger	17
Appendix D. Exact locked numerical record	17

1. Canonical locked statement

The Mass-Bridge is a **locked projective ratio system**, not a pair of separately fitted numerical correspondences. Define the proton-electron mass ratio η and the inverse fine-structure constant α^{-1} by

```
eta = m_p/m_e = 1836.152673426000
alpha^-1 = 137.035999177000
alpha = 1/alpha^-1
eta alpha = 13.399053419929
```

The authoritative constraints are

```
1000 A_bc/A_ab = eta = 1836.152673426000
100 A_bc/A_ac = alpha^-1 = 137.035999177000
10 A_ac/A_ab = eta alpha = 13.399053419929
```

The third equation is algebraically forced by the first two; it is not an additional fitted parameter. It also eliminates A_{bc} , making it the intrinsic Cell3 shape constraint. The source geometry reproduces the two primary ratios within approximately 8.6 parts per billion and the cross-ratio within 89.1 parts per trillion.

Equivalently, the exact face-area proportions are

$$A_{bc} : A_{ac} : A_{ab} = 1 : 100 \alpha : 1000/\eta$$

with the physical correspondence

$$A_{bc} : A_{ac} : A_{ab} \leftrightarrow \lambda_{\bar{e}} : 100 r_e : 1000 \lambda_{\bar{p}}$$

The factors 100 and 1000 are fixed registry placements within the Mass-Bridge definition. Only the overall common scale remains free. Fixing any one face area determines the other two. Retaining the native A_{bc} normalization gives

```
A_bc = 1.385784440909538e-03
A_ac = 1.011255764348181e-03
A_ab = 7.547217946337006e-04
```

Henceforth, the explicit η - and α -based ratios are authoritative. Previously reported face areas are numerical realizations of this locked triangle and are not independent adjustment variables. The three physical nodes represent the electron reduced-Compton inertial scale, the classical electron electromagnetic radius, and the proton reduced-Compton inertial scale.

The immediate geometric consequence is that the Cell3 face-area shape is forced. The bridge does not merely compare two ratios after the geometry has been selected. It selects the projective face-area point that the geometry must occupy.

2. Three physical scales form one closed triangle

The physical side of the bridge contains three lengths. They are not unrelated constants. They form a closed ratio triangle because all three are generated by the same electron and proton mass scales together with the electromagnetic coupling α .

```
lambda_bar_e = hbar/(m_e c)
lambda_bar_p = hbar/(m_p c)
r_e = alpha hbar/(m_e c) = alpha lambda_bar_e
```

Using the consistent CODATA 2022 values carried by the source record gives

Quantity	Definition	Value	Unit
electron reduced Compton wavelength	$\hbar/(m_e c)$	3.8615926744000e-13	m
classical electron radius	$\alpha \lambda_{\bar{e}}$	2.8179403204936e-15	m

Quantity	Definition	Value	Unit
proton reduced Compton wavelength	$\lambda_{\bar{e}}/\eta$	$2.1030891005348\text{e-}16$	m

Their exact ratio closure is

$$\lambda_{\bar{e}}/\lambda_{\bar{p}} = \eta = 1836.152673426000$$

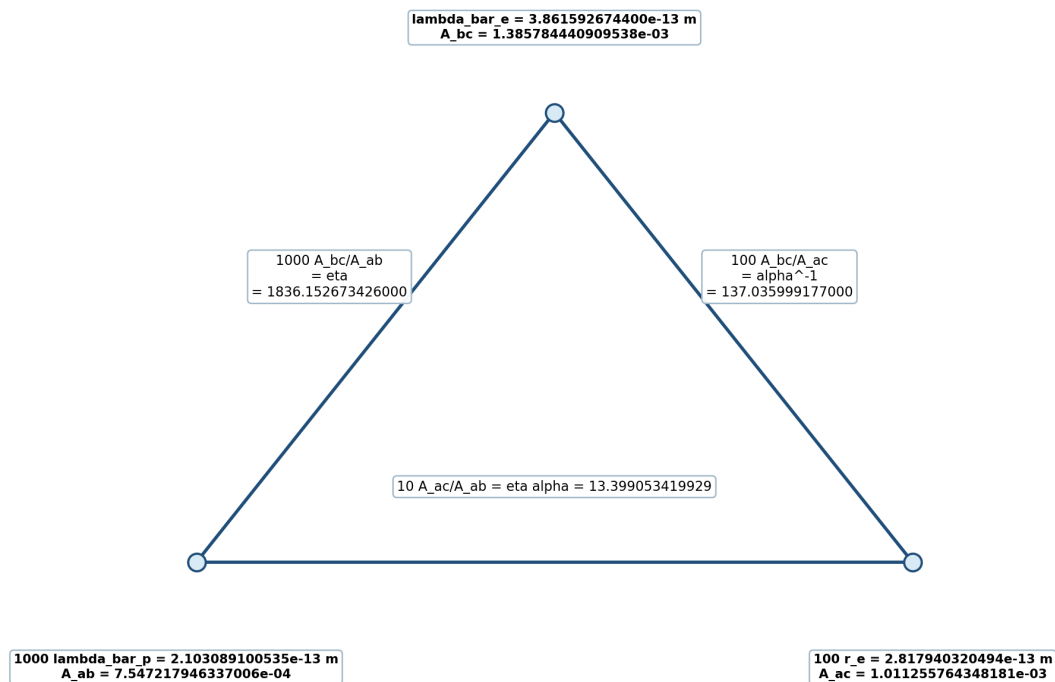
$$\lambda_{\bar{e}}/r_e = \alpha^{-1} = 137.035999177000$$

$$r_e/\lambda_{\bar{p}} = \eta \alpha = 13.399053419929$$

Multiplying the proton scale by 1000 and the classical-radius scale by 100 places the three physical lengths in the same projective registry as the three Cell3 face areas. The registry factors do not alter the underlying physics; they specify which decade slot each node occupies in the Mass-Bridge.

Compton-classical-radius Mass-Bridge triangle

Registered physical scales and exact locked Cell3 face areas



The factors 100 and 1000 are fixed registry placements; multiplication of all three areas by one positive scale leaves the triangle unchanged.

Figure 4. The closed Compton-classical-radius Mass-Bridge triangle. Each geometric face-area node is paired with one registered physical scale. The three edge labels are exact and mutually dependent.

2.1 Why the reduced Compton wavelength is the correct rotational scale

The reduced Compton wavelength uses $\hbar$ rather than h . That distinction is structural. Angular phase is measured in radians and closes after 2π radians, so the natural quantum of angular action is $\hbar$. The ordinary Compton wavelength contains the additional factor 2π and belongs to a linear cycle measured in full turns. The Mass-Bridge triangle is therefore a closed angular-phase construction.

$$\lambda_C = h/(m c) = 2\pi \lambda_{\bar{p}}$$

$$\text{closed angular phase} \leftrightarrow \hbar$$

$$\text{open linear wavelength} \leftrightarrow h$$

3. From three equations to two independent constraints

Although three ratio equations are written, the system contains only two independent dimensionless constraints. The third is generated by division. This dependence is exactly what makes the bridge closed.

$$\begin{aligned} (1000 \ A_{bc}/A_{ab}) &/ (100 \ A_{bc}/A_{ac}) \\ &= 10 \ A_{ac}/A_{ab} \\ &= \eta/(\alpha^{-1}) \\ &= \eta \alpha \end{aligned}$$

The cancellation of A_{bc} is decisive. Any common rescaling of the three face areas leaves all three bridge ratios unchanged, but the third ratio remains unchanged even if the common face A_{bc} is not referenced at all. It compares A_{ac} directly with A_{ab} and is therefore an internal shape test.

In ordinary vector space, the face-area triple is a point in the positive octant of R^3 . Projective geometry identifies triples that differ only by multiplication by a positive scalar. The Mass-Bridge equations select the single projective point

$$[A_{bc} : A_{ac} : A_{ab}] = [1 : 100 \alpha : 1000/\eta]$$

A common scale k produces the entire equivalent class

$$\begin{aligned} A_{bc} &= k \\ A_{ac} &= 100 \alpha k \\ A_{ab} &= (1000/\eta) k \\ \text{for any } k &> 0 \end{aligned}$$

This is the precise meaning of a locked projective ratio system. The lock fixes shape coordinates; it does not fix absolute size until one dimensional anchor is chosen.

4. Why the Cell3 shape becomes a forced constraint

A general parallelepiped has six intrinsic metric parameters: three edge lengths and three included angles. The Cell3 construction is not a general unconstrained parallelepiped. Its node grammar, basis orientation, face families, angular grammar, and existing metric realization are already established. Within that construction, the Mass-Bridge adds an exact projective condition on the three face channels. The combined system removes independent face-area freedom.

The forcing occurs in four layers. First, the address grammar fixes the combinatorial object: the 27 half-step nodes are $p_L = B L/2$ for L in $\{-1,0,1\}^3$. Second, the three primitive vectors fix the six plane families and their orientation. Third, the established included angles fix the shear class. Fourth, the Mass-Bridge fixes the relative magnitudes of the three cross products $b \times c$, $c \times a$, and $a \times b$. Once one face area is chosen as a scale anchor, all three face areas are numerically determined.

$$\begin{aligned} A_{bc} &= |b \times c| \\ A_{ac} &= |a \times c| \\ A_{ab} &= |a \times b| \end{aligned}$$

The source Cell3 already satisfies the lock to parts-per-billion precision. Therefore the exact locked realization is obtained by an extremely small correction rather than by a new shape search. The correction used here preserves the vector directions and all three included angles exactly. It changes only the primitive edge magnitudes by the minimum positive diagonal rescaling required to make the face areas exact.

The resulting statement is stronger than a numerical coincidence: the face ratios are boundary conditions on admissible Cell3 geometry. A candidate cell that does not occupy the locked projective point is not the Mass-Bridge Cell3.

5. Exact direction-preserving Cell3 correction

Let the source basis matrix B have primitive vectors a , b , and c as its columns. Apply positive scale factors s_a , s_b , and s_c without rotating the vectors:

```
B_locked = B_source diag(s_a, s_b, s_c)
a_locked = s_a a
b_locked = s_b b
c_locked = s_c c
```

The three face areas then transform as

```
A_bc_locked/A_bc_source = s_b s_c = r_bc
A_ac_locked/A_ac_source = s_a s_c = r_ac
A_ab_locked/A_ab_source = s_a s_b = r_ab
```

Taking logarithms gives a linear system. Its unique positive solution is

```
ln s_a = 1/2 (ln r_ac + ln r_ab - ln r_bc)
ln s_b = 1/2 (ln r_bc + ln r_ab - ln r_ac)
ln s_c = 1/2 (ln r_bc + ln r_ac - ln r_ab)
```

Because every scale factor is positive, the included angles are unchanged. Each face normal is also unchanged in direction because a cross product is multiplied only by a positive scalar. Thus the exact lock preserves the orientation, plane directions, and shear angles of the established Cell3.

Quantity	Source	Locked	Relative change (ppb)
s_a	1	0.99999999957983721	-0.042016279
s_b	1	0.99999999993699484	-0.006300516
s_c	1	1.00000000006300516	0.006300516
volume scale	1	0.99999999957983832	-0.042016168

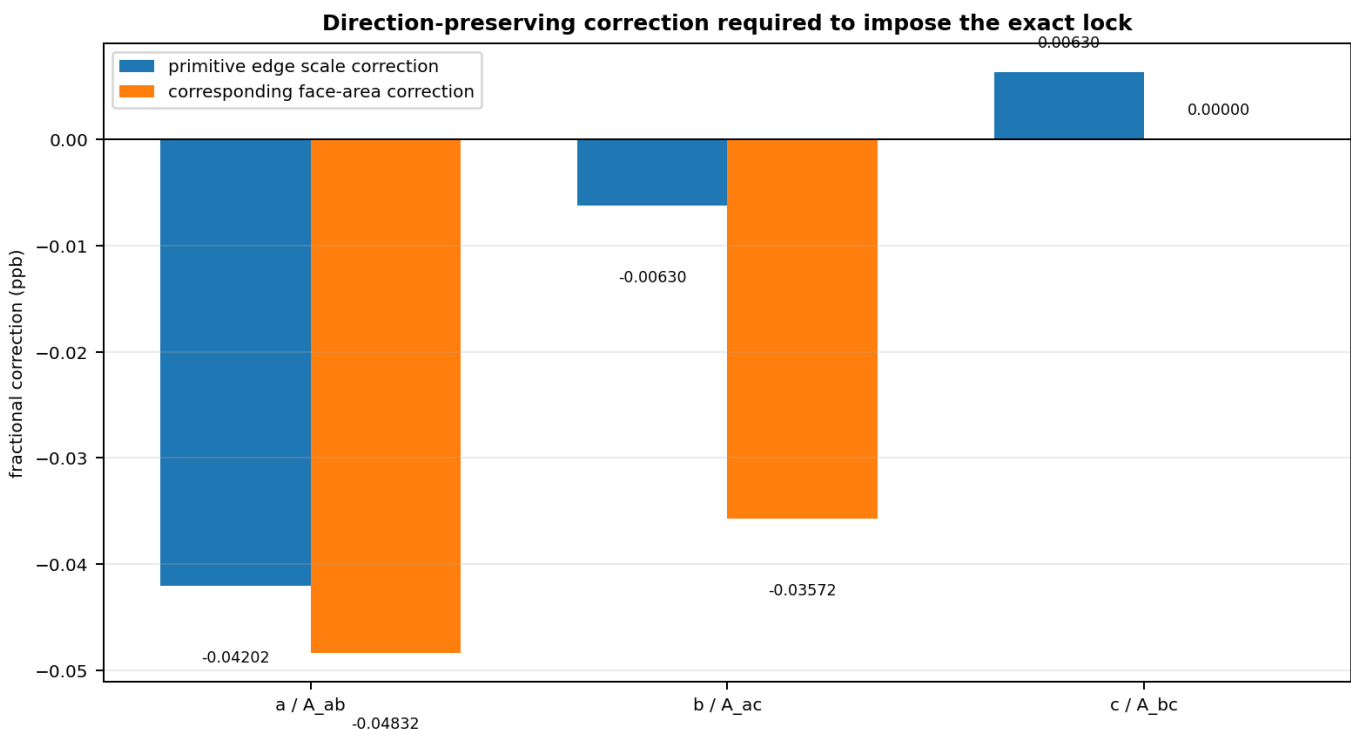


Figure 5. Primitive-edge and face-area corrections required to enforce the exact projective lock while retaining A_{bc} as the physical normalization anchor. The changes are tens of parts per trillion to tens of parts per billion.

5.1 Locked primitive basis

Vector	x (nm)	y (nm)	z (nm)	Length (nm)
a	0.327501242055891	-0.054897007884006	-0.037487502894730	0.334179679059111
b	0.035330775843828	0.207319850440192	-0.547401085388508	0.586410890411141
c	-0.010129317626729	-0.389787702366260	-0.689734408367268	0.792319765042685
Included angle	Locked value (deg)		Source value (deg)	Difference (deg)
angle(a, b)	83.93348301231273		83.93348301231273	0.000e+00
angle(a, c)	80.44813001833791		80.44813001833791	0.000e+00
angle(b, c)	50.36323181375618		50.36323181375619	-7.105e-15

6. Face-area closure and numerical comparison

Face channel	Source native area	Locked native area	Source physical area (nm^2)	Locked physical area (nm^2)
A_ab	7.547217880965119e-04	7.547217946337006e-04	0.194869165469900	0.194869165460484
A_ac	1.011255755679060e-03	1.011255764348181e-03	0.261106235799531	0.261106235790205
A_bc	1.385784440909538e-03	1.385784440909538e-03	0.357809539128562	0.357809539128562
Bridge quantity	Source geometry	Locked geometry	Physical target	Source relative residual
1000 A_bc/A_ab	1836.152689330240	1836.152673426000	1836.152673426000	8.661720 ppb
100 A_bc/A_ac	137.036000351759	137.035999177000	137.035999177000	8.572630 ppb
10 A_ac/A_ab	13.399053421123	13.399053419929	13.399053419929	0.089094 ppb

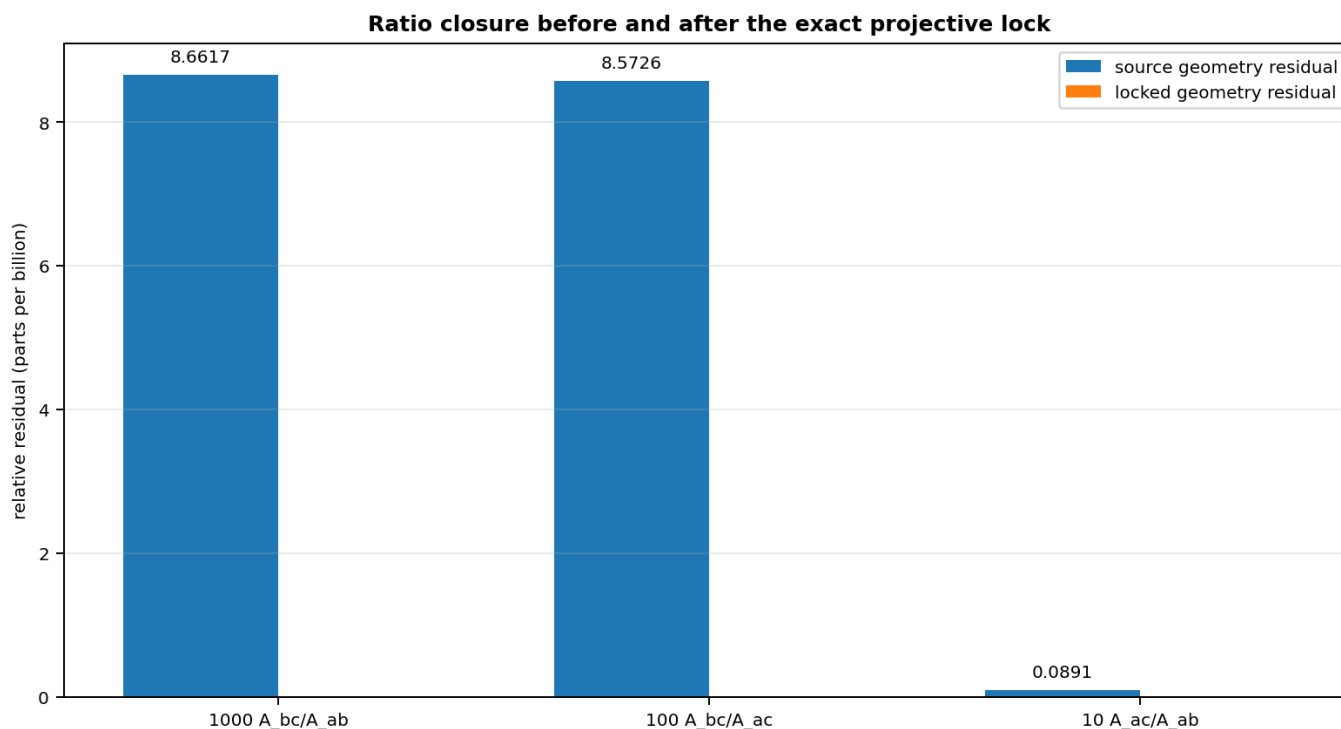


Figure 6. Source relative residuals expressed in parts per billion. The intrinsic cross-ratio residual is 0.0891 ppb, equivalent to 89.1 parts per trillion. The exact locked realization has zero residual at the displayed precision.

The source mismatch is coherent rather than random. Both primary bridge ratios are high by nearly the same fractional amount, while the shared-face-independent ratio is roughly one hundred times tighter. This pattern shows that most of the mismatch lies in a common normalization displacement associated with the shared face channel, not in a disagreement between the two independent shape directions.

7. Base-cell planes and metric data

The locked basis defines an oblique parallelepiped centered at the origin. Its six boundary planes occur at primitive coordinates $\pm 1/2$. In centered integer addresses $L=(i,j,k)$, the physical nodes are $p_L=B L/2$. The two bc faces have $i=\pm 1$; the two ac faces have $j=\pm 1$; and the two ab faces have $k=\pm 1$.

$$p_{(i,j,k)} = (i a + j b + k c)/2$$

$$(i,j,k) \text{ in } \{-1,0,1\}^3$$

Metric quantity		Value	
Cell volume		0.117914891911745 nm ³	
A _{bc}		0.357809539128562 nm ²	
A _{ac}		0.261106235790205 nm ²	
A _{ab}		0.194869165460484 nm ²	
opposite bc-face spacing		0.329546529695447 nm	
opposite ac-face spacing		0.451597379721285 nm	
opposite ab-face spacing		0.605097741518555 nm	
Face family	Unit normal x	Unit normal y	Unit normal z
bc	-0.995965190308961	0.083602162507377	-0.032619290564400
ac	-0.089052594101605	-0.866576769291032	0.491033948326276
ab	0.194092200807767	0.913177168821229	0.358379234790805

The diagonal correction leaves these normal directions unchanged. It also preserves the six plane incidence relations and the complete node membership of every plane. Only the metric distances and areas change by the minute locked corrections listed earlier.

8. The 27-node grammar remains exact

Cell3 contains eight corners, twelve edge centers, six face centers, and one body center. These are precisely the 27 elements of $\{-1,0,1\}^3$. The metric lock does not alter any address. It changes only the physical embedding B.

Therefore every parity rule, sheet assignment, adjacency relation, and antipodal pairing remains intact.

corner: 3 nonzero address components -> 8 nodes
 edge center: 2 nonzero components -> 12 nodes
 face center: 1 nonzero component -> 6 nodes
 body center: 0 nonzero components -> 1 node

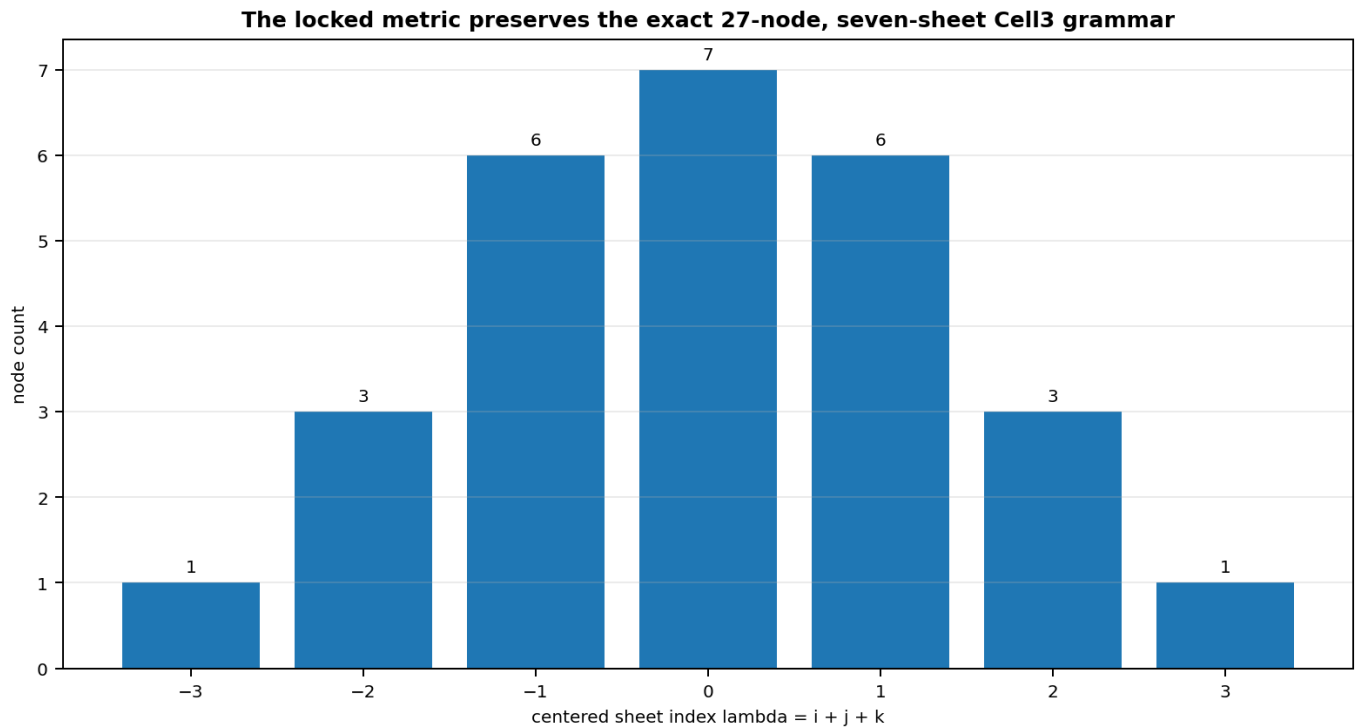


Figure 7. Node counts by centered sheet index $\lambda = i + j + k$. The exact 1,3,6,7,6,3,1 distribution is invariant under the locked metric correction.

The seven sheets are not extra geometric objects added to the cell. They are level sets of the integer address sum. Their preservation demonstrates that the forced shape acts at the metric layer while the finite Cell3 grammar remains exact.

9. Uniform scale is the only remaining geometric freedom

Once the projective face-area point and the established angular/orientation grammar are fixed, a uniform scale s maps the locked basis to sB . This transformation multiplies all edge lengths by s , all face areas by s^2 , and the volume by s^3 . Every Mass-Bridge ratio is unchanged.

```

B(s) = s B_locked
edge lengths(s) = s edge lengths(1)
face areas(s) = s^2 face areas(1)
volume(s) = s^3 volume(1)

```

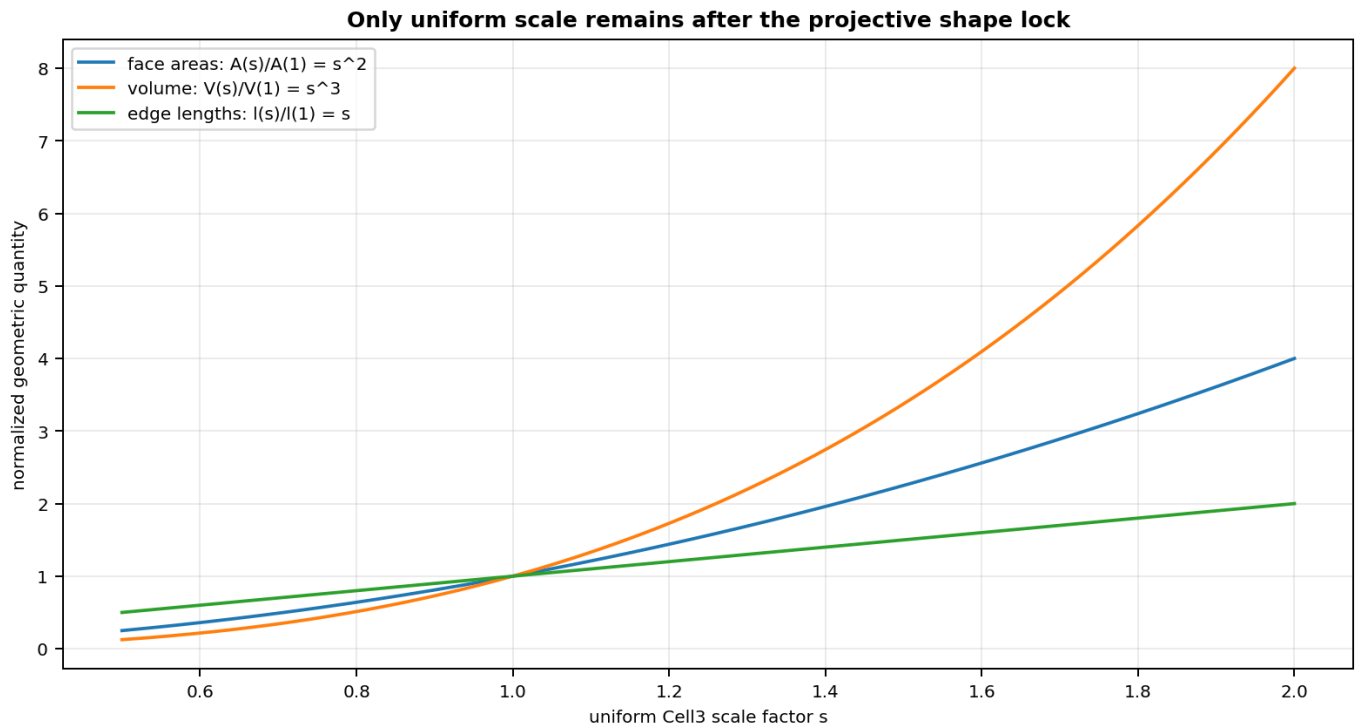


Figure 8. Exact dimensional scaling laws for the locked Cell3. The projective ratios remain constant along the entire curve because every face area carries the same s^2 factor.

This separation is essential for engineering and physical interpretation. Shape controls the dimensionless bridge ratios. Scale controls absolute edge lengths, areas, volume, and any scale-dependent interaction law. A prototype, molecular realization, or astrophysical realization can occupy a different dimensional scale while preserving the same Mass-Bridge Cell3 shape.

10. Consequences of the forced constraint

The locked system produces several direct consequences. These are mathematical consequences of the ratio and Cell3 definitions rather than additional fitted hypotheses.

First, independent face fitting is prohibited. Selecting A_{bc} and then fitting A_{ac} and A_{ab} separately would over-parameterize the cell. The locked equations determine both values. Any remaining discrepancy must be assigned to measurement, numerical representation, or a change of normalization rather than to independent bridge constants.

Second, the intrinsic cross-ratio is the primary shape witness. The quantity $10 A_{ac}/A_{ab}$ contains no A_{bc} . It therefore survives any uncertainty or recalibration confined to the shared face channel. Its source residual of 89.1 parts per trillion is the strongest direct evidence that the Cell3 shape already occupied the forced projective point before exact locking.

Third, the physical triangle replaces two unrelated coincidences. The mass ratio and inverse fine-structure ratio meet at the electron reduced Compton node. Their quotient is the classical-radius-to-proton-Compton ratio. The three relations are one closed object.

Fourth, physical registries become testable. The factors 100 and 1000 are part of the bridge definition. A different registry would move the Cell3 projective point and therefore produce a different forced shape. The present decade placement is not removable once the bridge is fixed.

Fifth, derivative models inherit a fixed carrier geometry. Any later K-Field calculation using Cell3 face channels, reciprocal metrics, plane spacings, or carrier tensors must use a basis consistent with the exact lock. The projective face-area ratios are upstream constraints, not downstream observables that may be retuned.

Sixth, data comparison becomes simpler. Experimental or computational Cell3 reconstructions can be reduced to two dimensionless residuals. One may compare $1000 A_{bc}/A_{ab}$ with η and $100 A_{bc}/A_{ac}$ with α^{-1} ; the cross-ratio then provides a dependent internal closure check.

11. Reconstruction protocol

The following procedure reconstructs the exact locked Cell3 from a source basis or from a single dimensional face anchor.

Step	Operation	Definition
1	Set the constants	$\eta=1836.152673426000$, $\alpha^{-1}=137.035999177000$, $\alpha=1/\alpha^{-1}$.
2	Choose one area anchor	Select $A_{bc}=k$ in the desired area unit.
3	Compute the other areas	$A_{ac}=100 \alpha k$ and $A_{ab}=(1000/\eta) k$.
4	If a source basis is available	Compute source face areas $ b \times c $, $ a \times c $, and $ a \times b $.
5	Form area correction ratios	$r_{bc}=A_{bc_locked}/A_{bc_source}$, $r_{ac}=A_{ac_locked}/A_{ac_source}$, $r_{ab}=A_{ab_locked}/A_{ab_source}$.
6	Solve the diagonal correction	Use the logarithmic half-sum formulas for s_a , s_b , and s_c .
7	Construct the locked basis	$B_{locked}=B_{source} \text{diag}(s_a, s_b, s_c)$.
8	Generate all nodes	For every L in $\{-1, 0, 1\}^3$, compute $p_L=B_{locked} L/2$.
9	Generate planes	Use $i=\pm 1$ for bc planes, $j=\pm 1$ for ac planes, and $k=\pm 1$ for ab planes.
10	Verify closure	Recompute all three ratios, included angles, plane normals, node counts, and volume.

The reconstruction is deterministic. Given the constants, an area anchor, and the established Cell3 directions, no fitting loop remains.

12. Formula ledger

Object	Equation
Physical scales	$\lambda_{\bar{e}}=\hbar/(m_e c)$; $\lambda_{\bar{p}}=\hbar/(m_p c)$; $r_e=\alpha \lambda_{\bar{e}}$
Physical ratios	$\lambda_{\bar{e}}/\lambda_{\bar{p}}=\eta$; $\lambda_{\bar{e}}/r_e=\alpha^{-1}$; $r_e/\lambda_{\bar{p}}=\eta \alpha$
Face areas	$A_{bc}= b \times c $; $A_{ac}= a \times c $; $A_{ab}= a \times b $
Locked bridge	$1000 A_{bc}/A_{ab}=\eta$; $100 A_{bc}/A_{ac}=\alpha^{-1}$; $10 A_{ac}/A_{ab}=\eta \alpha$
Projective point	$[A_{bc}:A_{ac}:A_{ab}]=[1:100 \alpha:1000/\eta]$
Node embedding	$p_L=B L/2$, L in $\{-1, 0, 1\}^3$
Direct metric	$M=B^T B$
Reciprocal metric	$R=M^{-1}$
Volume	$V= \det B $
Plane spacing	$h_a=V/A_{bc}$; $h_b=V/A_{ac}$; $h_c=V/A_{ab}$
Uniform scaling	$B \rightarrow sB$; $A \rightarrow s^2 A$; $V \rightarrow s^3 V$
Diagonal correction	$B_{locked}=B_{source} \text{diag}(s_a, s_b, s_c)$

13. Data provenance and reproducibility contract

All figures and numerical tables in this report are generated from two supplied records: the attached Mass-Bridge analysis and the authoritative Cell3 machine-readable source. The physical constants are the CODATA 2022 values carried by the attached analysis and the SciPy 1.17.0 physical-constants table. No coordinates, nodes, plane memberships, areas, or residuals were visually estimated.

Source	Role	SHA-256
Pasted text(145).txt	Mass-Bridge physical triangle, native face areas, constants, and source ratio comparison	33b8359e07a9f31ebb523822a6c2b5727e2b7fb08fbc1be836b1bd1569f4bb81
K-Field_single_cell3_lateral_force_reduction_1784985105.json	Authoritative Cell3 basis, metric, face areas, plane families, and node grammar	62e4472aacc7df503cc4b5120d7abce97589891e8d22843177ee547e35b71a6f

The companion JSON file records the full locked basis, metrics, areas, ratios, nodes, plane definitions, source hashes, figure hashes, and reconstruction equations. This makes the report independently reproducible without extracting values from page graphics.

Appendix A. Locked matrices

Locked basis B (columns are a, b, c)

B (nm)	column 1	column 2	column 3
row 1	3.275012420558906e-01	3.533077584382840e-02	-1.012931762672882e-02
row 2	-5.489700788400644e-02	2.073198504401918e-01	-3.897877023662599e-01
row 3	-3.748750289472991e-02	-5.474010853885081e-01	-6.897344083672676e-01

Locked direct metric $M=B^T B$

M (nm ²)	column 1	column 2	column 3
row 1	1.116760578960501e-01	2.071033328060581e-02	4.393723509621977e-02
row 2	2.071033328060581e-02	3.438777323927870e-01	2.963927589615229e-01
row 3	4.393723509621977e-02	2.963927589615229e-01	6.277706100772958e-01

Locked reciprocal metric $R=M^{-1}$

R (nm ⁻²)	column 1	column 2	column 3
row 1	9.208025529457549e+00	1.534802428192545e-03	-6.451880356999390e-01
row 2	1.534802428192587e-03	4.903398312367987e+00	-2.315175584416389e+00
row 3	-6.451880356999389e-01	-2.315175584416389e+00	2.731171274711695e+00

Appendix B. Complete boundary-plane ledger

Plane	Family	Primitive coordinate	Normal x	Normal y	Normal z	Area (nm ²)	Opposite spacing (nm)
a-1	bc	-0.5	-0.995965190309	0.083602162507	-0.032619290564	0.357809539129	0.329546529695
a+1	bc	+0.5	-0.995965190309	0.083602162507	-0.032619290564	0.357809539129	0.329546529695
b-1	ac	-0.5	-0.089052594102	-0.866576769291	0.491033948326	0.261106235790	0.451597379721
b+1	ac	+0.5	-0.089052594102	-0.866576769291	0.491033948326	0.261106235790	0.451597379721
c-1	ab	-0.5	0.194092200808	0.913177168821	0.358379234791	0.194869165460	0.605097741519
c+1	ab	+0.5	0.194092200808	0.913177168821	0.358379234791	0.194869165460	0.605097741519

Each plane contains the nine addresses for which its corresponding address component equals +1 or -1. The full address membership is shown graphically in Figure 3 and stored explicitly in the companion JSON.

Appendix C. Complete 27-node locked Cell3 ledger

Address (i,j,k)	Kind	Sheet lambda	Polarity	x (nm)	y (nm)	z (nm)
(-1, -1, -1)	corner	-3	-1	-0.176351350136495	0.118682429905037	0.637311498325253
(-1, -1, 0)	edge center	-2	1	-0.181416008949859	-0.076211421278093	0.292444294141619
(-1, 0, -1)	edge center	-2	1	-0.158685962214581	0.222342355125133	0.363610955630999
(0, -1, -1)	edge center	-2	1	-0.012600729108550	0.091233925963034	0.618567746877888
(-1, -1, 1)	corner	-1	-1	-0.186480667763224	-0.271105272461223	-0.052422910042015
(-1, 0, 0)	face center	-1	-1	-0.163750621027945	0.027448503942003	0.018743751447365
(-1, 1, -1)	corner	-1	-1	-0.141020574292667	0.326002280345229	0.089910412936745
(0, -1, 0)	face center	-1	-1	-0.017665387921914	-0.103659925220096	0.273700542694254
(0, 0, -1)	face center	-1	-1	0.005064658813364	0.194893851183130	0.344867204183634
(1, -1, -1)	corner	-1	-1	0.151149891919396	0.063785422021031	0.599823995430523
(-1, 0, 1)	edge center	0	1	-0.168815279841310	-0.167445347241127	-0.326123452736269
(-1, 1, 0)	edge center	0	1	-0.146085233106031	0.131108429162099	-0.254956791246889
(0, -1, 1)	edge center	0	1	-0.022730046735279	-0.298553776403226	-0.071166661489380
(0, 0, 0)	body center	0	1	0.000000000000000	0.000000000000000	0.000000000000000
(0, 1, -1)	edge center	0	1	0.022730046735279	0.298553776403226	0.071166661489380
(1, -1, 0)	edge center	0	1	0.146085233106031	-0.131108429162099	0.254956791246889
(1, 0, -1)	edge center	0	1	0.168815279841310	0.167445347241127	0.326123452736269
(-1, 1, 1)	corner	1	-1	-0.151149891919396	-0.063785422021031	-0.599823995430523
(0, 0, 1)	face center	1	-1	-0.005064658813364	-0.194893851183130	-0.344867204183634
(0, 1, 0)	face center	1	-1	0.017665387921914	0.103659925220096	-0.273700542694254
(1, -1, 1)	corner	1	-1	0.141020574292667	-0.326002280345229	-0.089910412936745
(1, 0, 0)	face center	1	-1	0.163750621027945	-0.027448503942003	-0.018743751447365
(1, 1, -1)	corner	1	-1	0.186480667763224	0.271105272461223	0.052422910042015
(0, 1, 1)	edge center	2	1	0.012600729108550	-0.091233925963034	-0.618567746877888
(1, 0, 1)	edge center	2	1	0.158685962214581	-0.222342355125133	-0.363610955630999
(1, 1, 0)	edge center	2	1	0.181416008949859	0.076211421278093	-0.292444294141619
(1, 1, 1)	corner	3	-1	0.176351350136495	-0.118682429905037	-0.637311498325253

Appendix D. Exact locked numerical record

Field	Value
eta	1836.152673426000
alpha^-1	137.035999177000
alpha	7.297352564331424e-03
eta alpha	13.399053419929
native A_bc anchor	1.385784440909537977e-03
native A_ac locked	1.011255764348180635e-03
native A_ab locked	7.547217946337006099e-04
physical A_bc anchor	3.578095391285615046e-01 nm^2
physical A_ac locked	2.611062357902053099e-01 nm^2
physical A_ab locked	1.948691654604841339e-01 nm^2
volume	1.179148919117446459e-01 nm^3
report epoch	1785176592
generation UTC	2026-07-27T18:23:12Z